Portfolio
Hardware prototyping

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Hardware prototyper

I design, build and document electromechanical devices end to end: CAD, PCB layout, 3D-printed enclosures, wire harnesses, soldering and embedded firmware. These four projects range from a test fixture built on a production floor to handheld instruments.

Tools KiCad · Fusion 360 · OpenSCAD · Rhino + Grasshopper · Blender · FDM printing · laser cutting · soldering · crimping · C/C++ firmware



01 · P.2-3

Wire harness tester

Bench tester for Molex Micro-Fit+ harnesses. It went from a perfboard prototype to a 10-unit pilot build.

Tesla (contract) · 2026

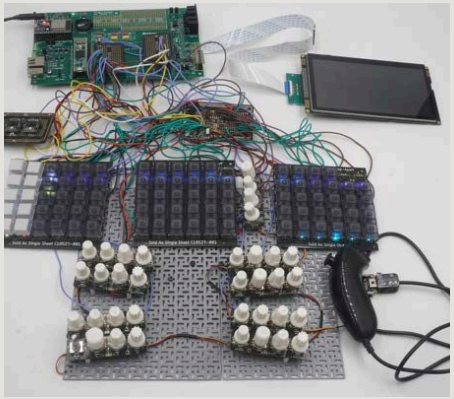


02 · P.4

Tombogo × Rheome

Handheld audio instrument: ESP32-S3, round touch display, custom daughterboard, nine enclosure revisions.

Rheome · 2025-26



03 · P.5

Mutax

Gesture-controlled performance instrument on a Teensy 4.1, hand-wired across key matrices and encoder banks.

NYU Tisch thesis · 2024-25



04 · P.6

kymecreo

Velocity-sensitive split MIDI keyboard using hall-effect switches. The enclosure was sculpted, scanned and remodeled in CAD.

Rheome · 2024

Test fixture · built on my own initiative

Wire harness tester

I built a bench tester, out of my own parts, that catches wiring faults in harnesses before they leave the line.

2	3	10
faulty harnesses caught in its first batch	hardware revisions in my first 3 weeks	units in the pilot build

What one plug-in checks

- **Continuity, opens and shorts** on every wire
- **Crossed pins:** a wire landing on the wrong position
- **Resistance** against a zeroed, known-good harness. It fails anything more than 3 Ω over baseline
- **Intermittent crimps:** wiggle the harness while it's plugged in and any dropout is flagged

Using it

Plug in both ends. The screen shows the four positions as they sit on the connector face, and the tester beeps pass or fail. Settle and hold timers keep the act of plugging in from registering as a failure. The housings on each side are wired in parallel, so it accepts either harness end.

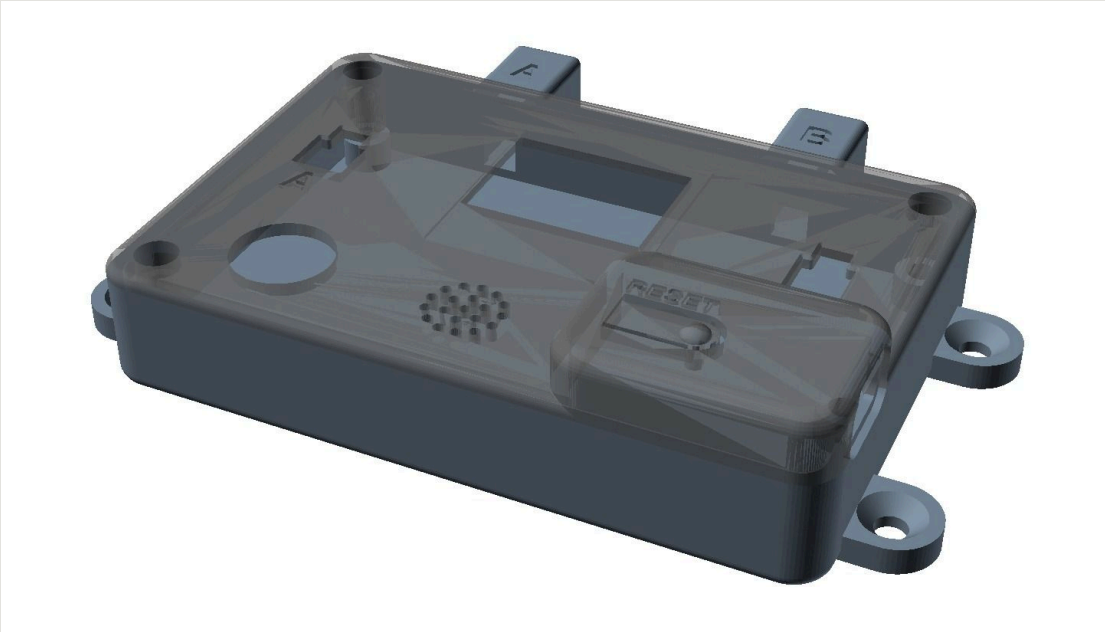


Fig. 1 · Parametric OpenSCAD enclosure: A/B connector ports, flush OLED window, buzzer grille, wall-mount ears

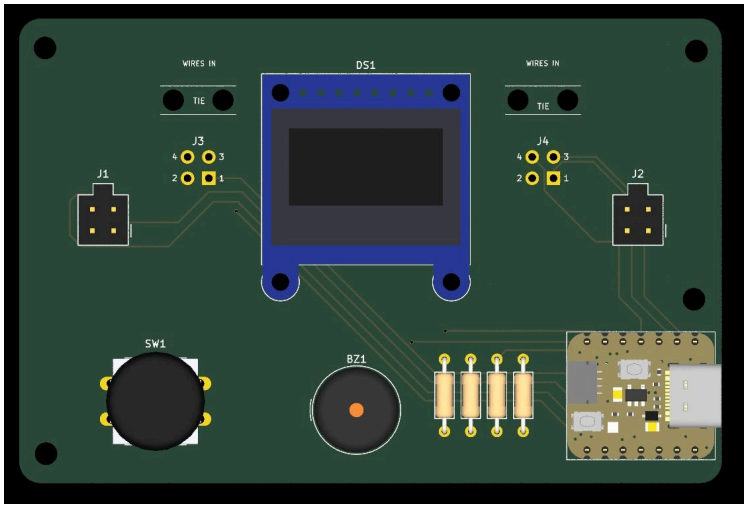
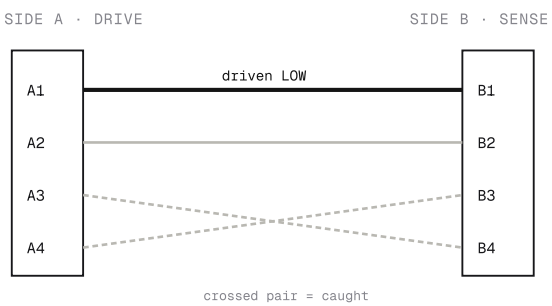


Fig. 2 · 2-layer PCB, 99 × 65 mm, laid out in KiCad: QT Py RP2040, Micro-Fit+ headers, pigtail pads with cable-tie strain relief

Rev A Late Sep 2026

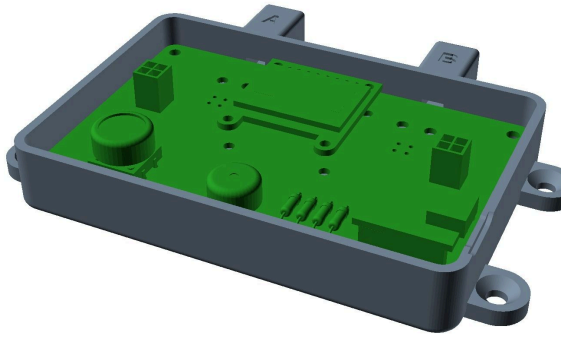
Proof of concept



- Arduino Nano 33 IoT and an SSD1306 OLED on perfboard, built from parts I had on hand.
- **Method:** drive one A pin low while the others float. Each B pin has a 4.7 kΩ pull-up into an ADC, so whichever B pins drop show where that wire lands, and the voltage gives its resistance. A single scan catches opens, shorts and crossed pins.

Rev B Oct 2026

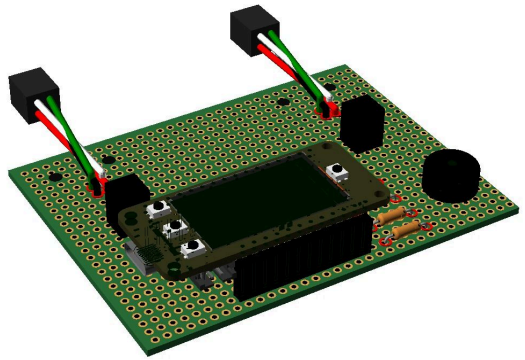
Custom PCB + enclosure



- QT Py RP2040 on a KiCad PCB. I generated the project, libraries and 3D models from Python scripts so I could regenerate them after each change.
- Parametric OpenSCAD enclosure with a back-wall pigtail port, heat-set inserts, a printed button cap, and the PCB's STEP export as the fit reference.
- Produced fab-ready Gerbers, and an EE looked over the layout.

Rev C Oct 2026 · ordering

10-unit pilot build



- I recommended protoboard over the PCB for the pilot. That way the tester proves its value before anyone spends engineering review time on a board.
- Moved to an ESP32-S3 Reverse TFT Feather with the display and buttons onboard. That cut the part count for hand assembly.
- Laid out the protoboard in KiCad on a 0.1 in grid and handed off a 10-unit DigiKey order.

Build package delivered with each revision

- Gerbers + drill files, zipped for any fab
- BOM with DigiKey part numbers, unit price and stock
- STL print files with orientation and support notes
- Pigtail crimping instructions (Micro-Fit+ 206460)
- Step-by-step assembly guide
- Firmware, pin map and flashing steps
- Calibration: zero against a known-good harness
- Open issues list, so nothing is hidden

Next

Mount testers along the line so every harness is checked as part of the build.

Handheld audio instrument

Tombogo × Rheome handheld

A pocket instrument with a retroactive looper, a resonator, a chopper and a drum machine, all played on a round touch display. It's scheduled for a Kickstarter launch in December 2026.

What I built

- **Daughterboard (KiCad):** through-hole board with an MCP23017 I/O expander, rotary encoder, 5-way nav switch, and a switched headphone jack with output conditioning. It went through two revisions with DRC passes and a written design review. The production mainboard is now in placement.
- **Enclosure:** modeled in OpenSCAD and rendered in Blender, over nine revisions so far. They added battery-wire access, connector relief, seam closure and button fit.
- **Firmware:** C/C++ on an ESP32-S3 with a 466 × 466 AMOLED, IMU and onboard mics. DSP is in Faust and the UI in LVGL.
- **Process:** 600+ commits. Every feature you can hear, feel or see has to pass a hands-on check on the powered device before it counts as done.

MCU	ESP32-S3, 8 MB PSRAM, 16 MB flash
Display	1.75 in round AMOLED, capacitive touch
I/O	Encoder, nav switch, 3 side buttons, IMU, mics, headphone out
Tools	KiCad · OpenSCAD · Blender · Faust · LVGL · PlatformIO



Fig. 3 · Working prototype, chord UI

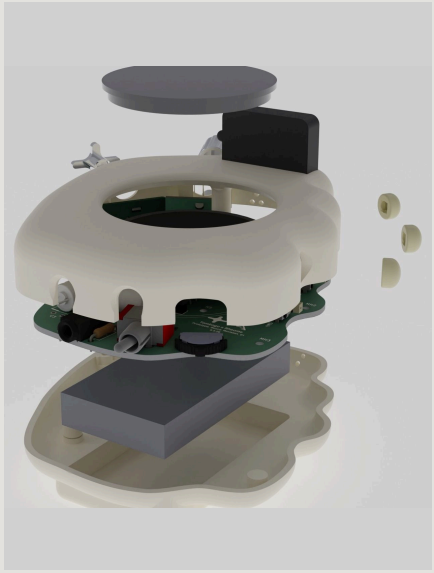


Fig. 4 · Exploded assembly render

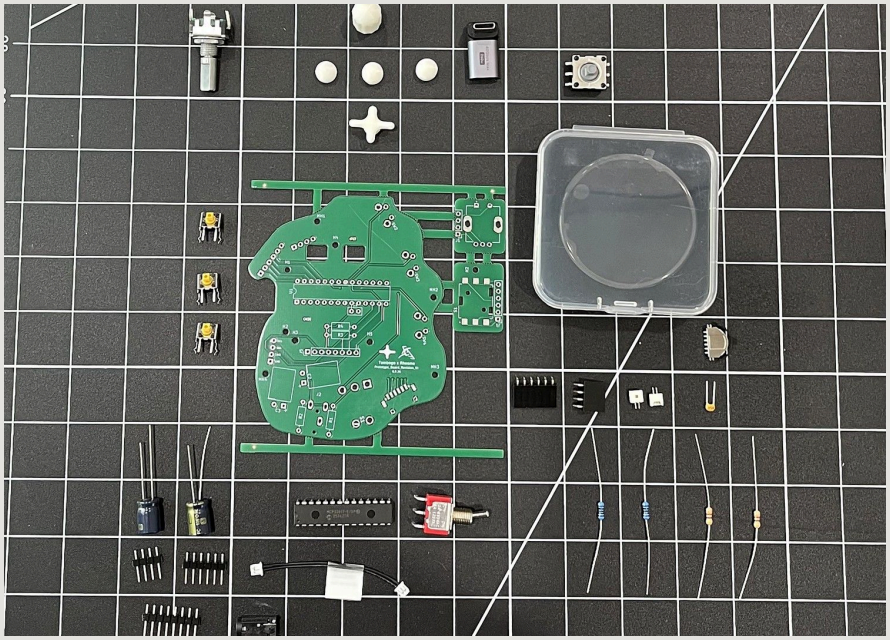


Fig. 5 · Rev 01 daughterboard back from fab, with the encoder, switches, printed caps and passives laid out before assembly

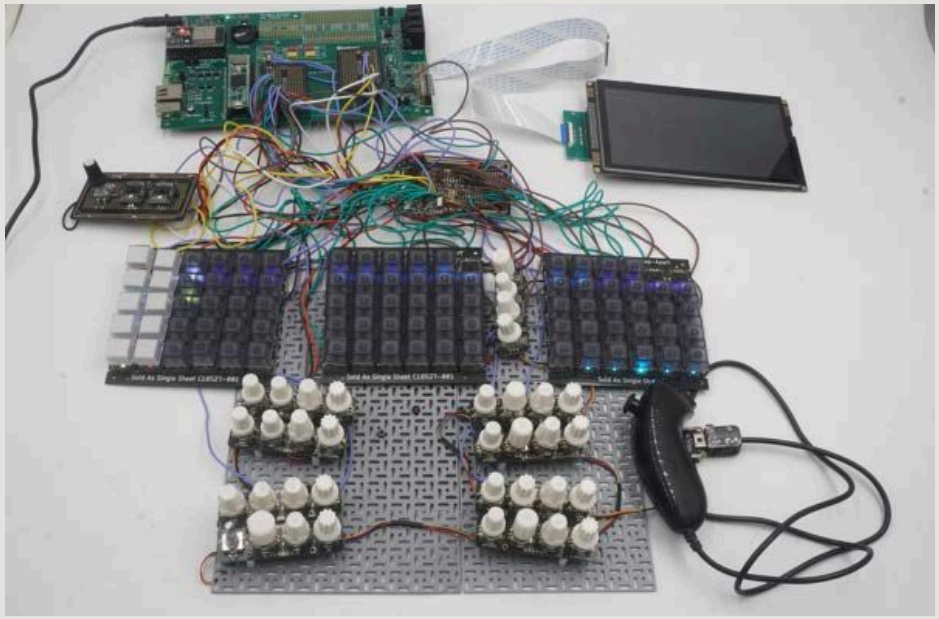


Fig. 6 · Thesis rig: Teensy 4.1 baseboard, 3 key matrices, 9 quad-encoder boards, Nunchuk gesture input

Looper

Every note is timestamped into a PSRAM ring buffer. Hit a length key and the last few bars loop, with no need to arm recording.

Sequencer

Euclidean pattern per track, plus a distribution parameter that pushes hits toward the front or back of the bar.

Change engine

A menu of harmonic moves (scale degree, chord type) and energy moves (build, drop, breakdown).

Gesture

Hold Z: the accelerometer plays fills and the joystick shifts octaves. Release: the cued change lands on every track.

Brain	Teensy 4.1 with PSRAM, internal clock or external MIDI sync
Controls	3× 5×6 key matrices · 9 quad encoder boards · Wii Nunchuk (I²C)

Performance instrument

Mutax

An instrument for improvising live electronic music. One player can shift rhythm, harmony and energy across every track at once, with a hand gesture.

- **Spec from field notes:** I improvised for 20 minutes a day and logged every moment the existing gear got in the way. That log became the requirements.
- **Hardware:** scaled up to three key matrices on three scanner ICs and nine quad-encoder boards, all hand-soldered across two perfboards.
- **Debugging:** I tracked down I²C ghosting caused by a matrix split across two scanners, and encoder boards reporting addresses that changed overnight.
- **Porting:** moved my browser Euclidean sequencer from JavaScript to C++ for the microcontroller.
- **PCB:** consolidated the controls into a sub-module board (below) with 74HC4051 analog multiplexers and SK6812 addressable LEDs, then exported Gerbers, BOM and placement files for JLCPCB assembly.

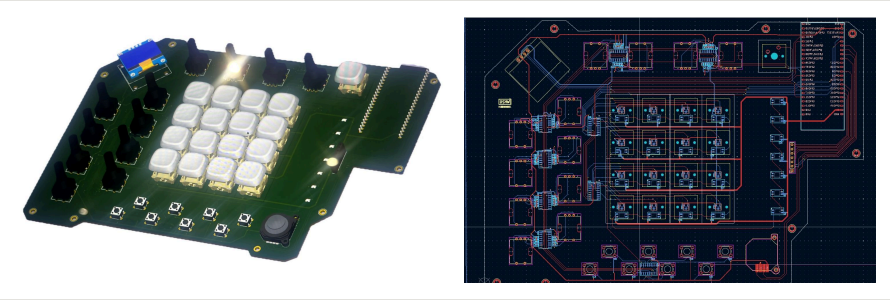


Fig. 7 · Sub-module PCB in KiCad: 4×4 keybed, 12 encoders, 8 buttons, joystick, OLED

Watch

Live improvisation and thesis talk: youtube.com/watch?v=Fc16EAlU_OQ
Build log: rheome.tools/instruments/mutax

Velocity-sensitive MIDI controller

kymecreo

An ergonomic split MIDI keyboard with hall-effect switches. Each key reports how far it's pressed, not just whether it's down, so it can send velocity.

- **Proof of concept:** one analog magnetic switch over an OH49E hall sensor, wired into an Arduino Nano. The serial plotter showed a clean analog curve for key depth.
- **Layout by hand:** I put my hand on the perfboard and marked where my fingers landed, then soldered the sensor array in two nights. The joystick got its own sub-board so power and ground couldn't short on a shared trace.
- **Debugging:** one key read high at rest. It sat under a 5 V trace, and the trace's field was reaching its hall sensor.
- **Aftertouch:** a silicone O-ring inside the keycap gives extra travel to read after the key bottoms out.

Print	Sculpt	Scan	Model
Printed the open-source Skeletyl split and cut off the thumb cluster	Built Sculpey around it to fit my hand and the joystick	3D-scanned the clay with an iPad	Remodeled it in Fusion 360 with the scan as reference

Switches	Analog hall-effect magnetic (Lekker / Gateron)
Brain	Teensy 4.1, USB MIDI
Extra	Joystick as an XY thumb mod wheel

Watch

Demo: youtu.be/md1QRLlQi-M · Build log: rheome.tools/archive/kymecreo



Fig. 8 · Printed enclosure, modeled in Fusion 360 from the scanned clay sculpt

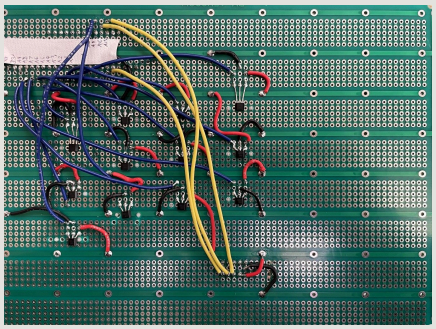


Fig. 9 · Hand-wired hall sensor array

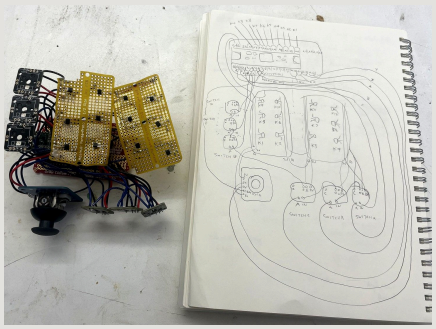


Fig. 10 · Sub-boards and the wiring map I drew first